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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Cars24 Used Car Market Analysis
Using Web Scraping & Exploratory Data Analysis

About me

Name : Kunchapu Nandusree

Background : B Tech (Information Technology)

Project : Cars24 Used Cars Data Analysis – Web Scrapping & Data Analytics

LinkedIn : https://www.linkedin.com/in/nandu-kunchapu-b02210321/

GitHub :https://github.com/kunchapunandu18-dev

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Agenda

Business Problem
Project Objective
Tools & Technologies
Web Scraping & Data Collection
Dataset & Data Preprocessing
Exploratory Data Analysis & Visualization
Statistical & Descriptive Analysis
Key Business Insights
Conclusion

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Business Problem

Cars24 provides used cars across multiple brands, model years, fuel types, transmission types and price points. With many listings available, buyers may find it difficult to compare vehicles and understand what drives resale prices.

This project collects used-car listing data from Cars24 and analyzes price, kilometers, fuel type, transmission, model year, EMI and brand to uncover market patterns and support more informed purchasing decisions.

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Project Objective

- Scrape used-car listings from Cars24 across multiple brands.
- Clean and transform raw price, EMI, kilometer and year fields.
- Extract brand information from vehicle names.
- Analyze price patterns by brand, fuel type, transmission and model year.
- Study the relationship between kilometers driven and price.
- Identify high-value patterns, outliers and key market insights.
- Prepare a clean dataset for further analysis and reporting.

Tools & Technologies

1. **Python** – Programming & analysis
2. **Requests** – Web data collection
3. **BeautifulSoup** – Web scraping
4. **Pandas** – Data cleaning & manipulation
5. **NumPy** – Numerical analysis
6. **Matplotlib & Seaborn** – Data visualization
7. **SciPy** – Statistical & hypothesis testing
8. **Jupyter Notebook** – Development & analysis environment

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Web Scraping & Data Collection

SOURCE

Cars24 used-car listing pages

SCRAPING APPROACH

17 brands × 2 pages per brand

Requests + BeautifulSoup + Regex

DATA COLLECTED

- 293 unique car records
- Car name and brand
- Kilometers, fuel type and transmission
- EMI, price and model year

The extracted records were stored in a Pandas DataFrame and prepared for cleaning and analysis.

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Dataset & Data Preprocessing

Column	Data Type	Missing Values	Missing %	Unique Values
Car_Name	object	0	0.0	267
Kilometers	object	0	0.0	293
Fuel_Type	object	0	0.0	3
Transmission	object	0	0.0	2
EMI	object	0	0.0	288
Price	object	0	0.0	257
Model_Year	int64	0	0.0	14
Price_Lakh	float64	0	0.0	257
EMI_Amount	float64	1	0.34	287
Brand	object	0	0.0	17

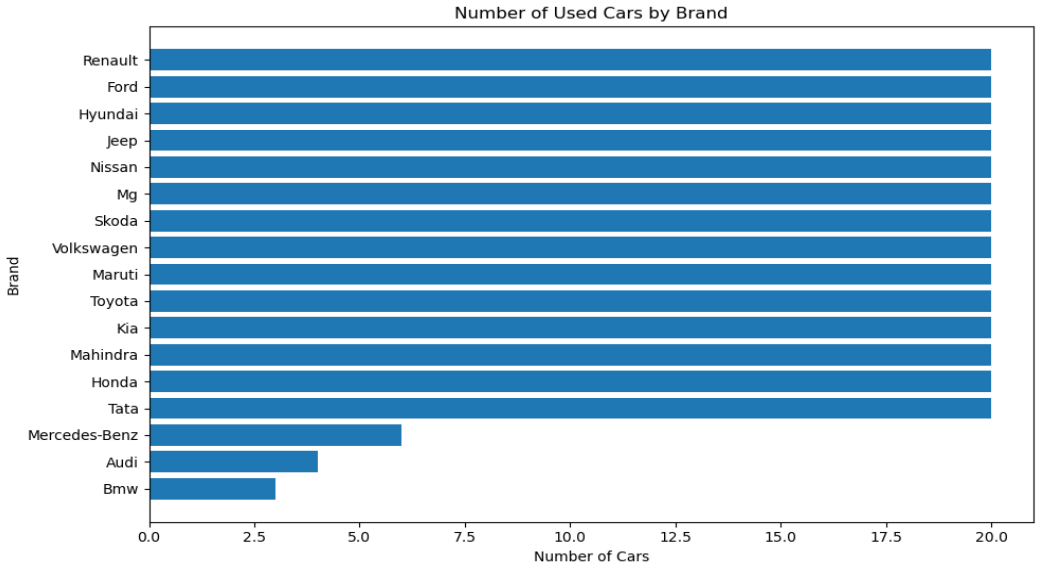
DATASET OVERVIEW

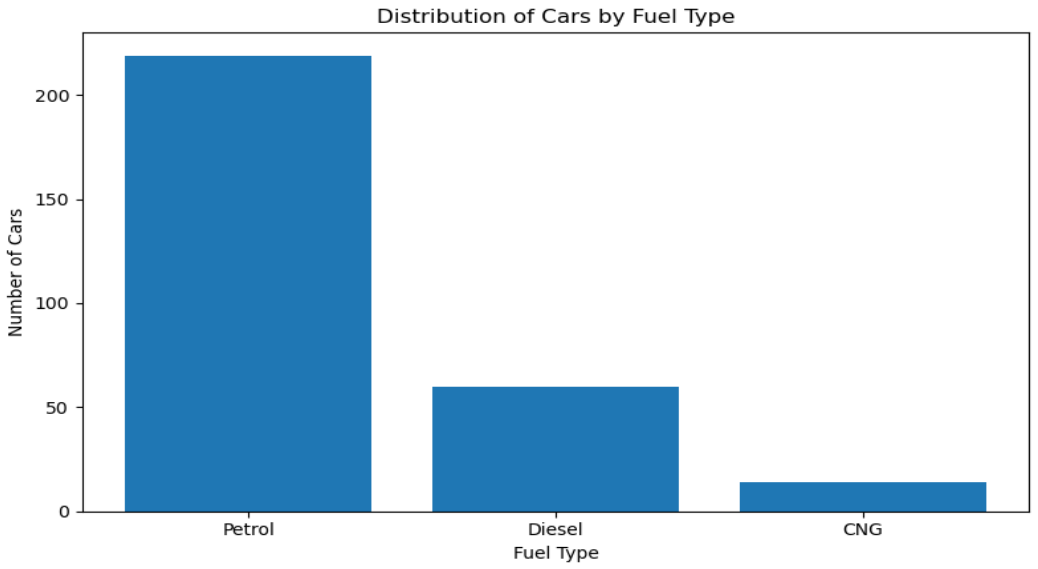
- Initial dataset: 293 records
- Final dataset: 293 rows × 10 columns
- 17 brands, 3 fuel types and 2 transmission types
- Model years span 2013–2026

PREPROCESSING

- Converted price and EMI to numeric values
- Extracted brand from car names
- Converted model year to integer
- Checked duplicates and missing values
- Created kilometer ranges for comparison
- Identified price outliers using the IQR method

Exploratory Data Analysis





BRAND PRESENCE

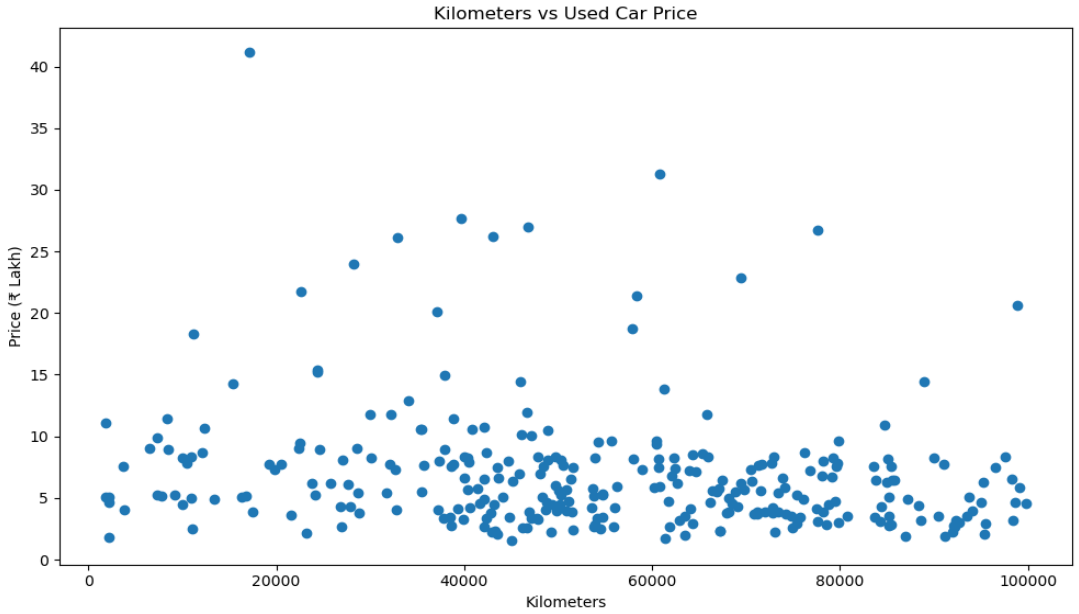
Most brands contribute 20 listings each; Mercedes-Benz, Audi and BMW have fewer records because of the scraped sample.

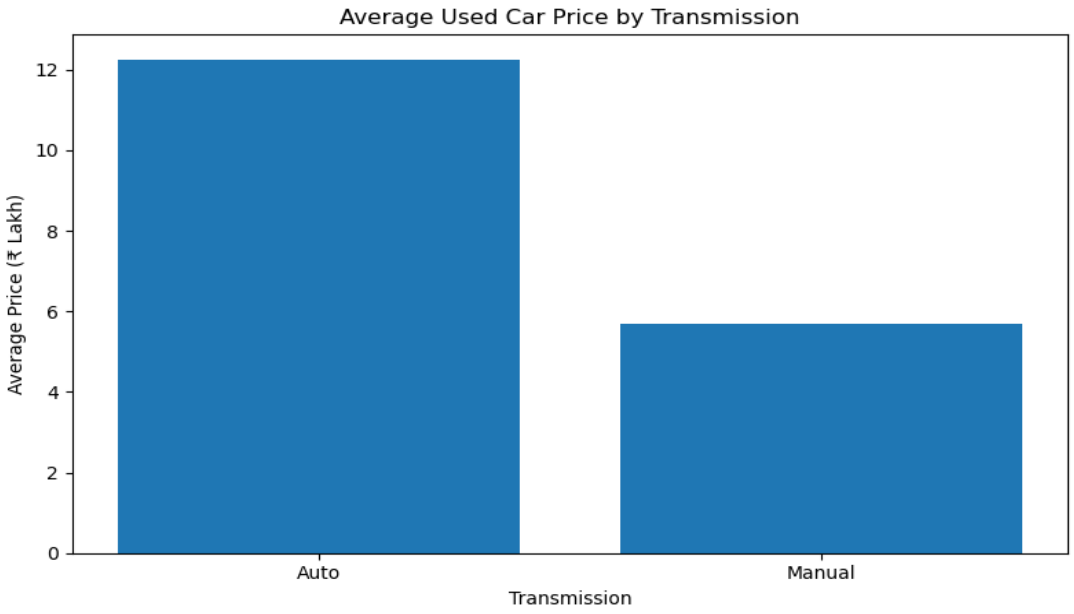
FUEL MIX

Petrol dominates the dataset with 219 cars, followed by Diesel (60) and CNG (14).

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Bivariate Analysis





KILOMETERS VS PRICE

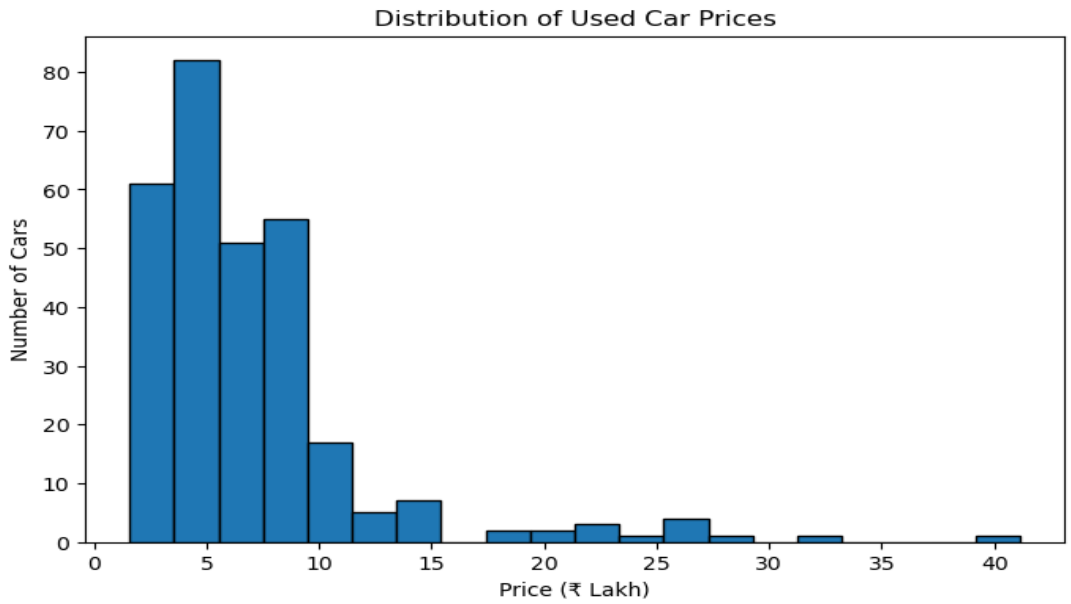
Price has a weak negative relationship with kilometers driven (correlation ≈ −0.19). Higher mileage is associated with somewhat lower prices, but the relationship is not strong.

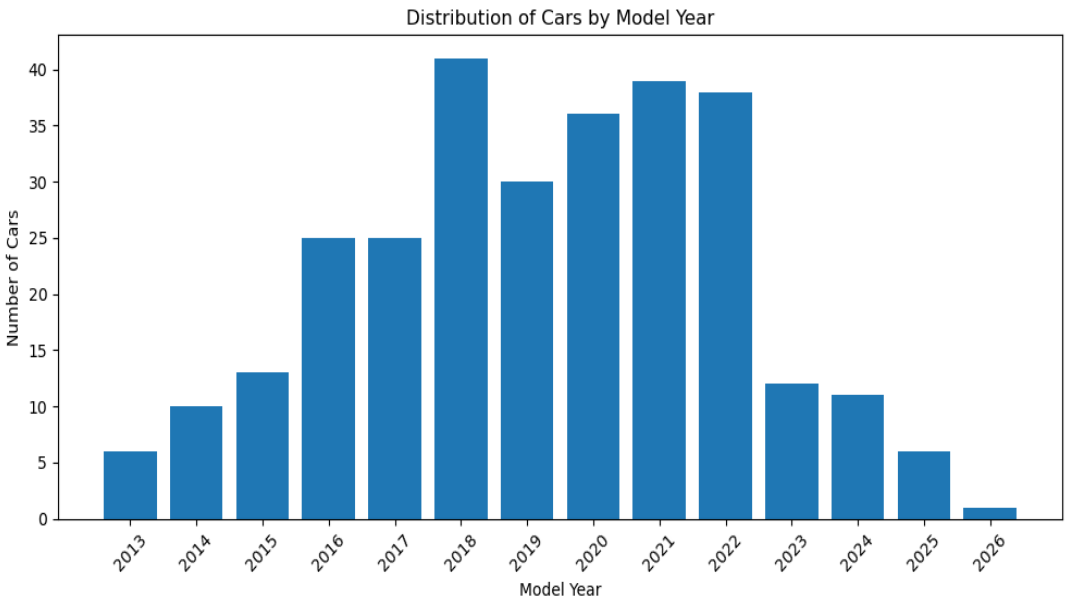
TRANSMISSION VS PRICE

Automatic cars have a higher average price (about ₹12.26 lakh) than manual cars (about ₹5.69 lakh) in this sample.

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PRICE DISTRIBUTION





PRICE DISTRIBUTION

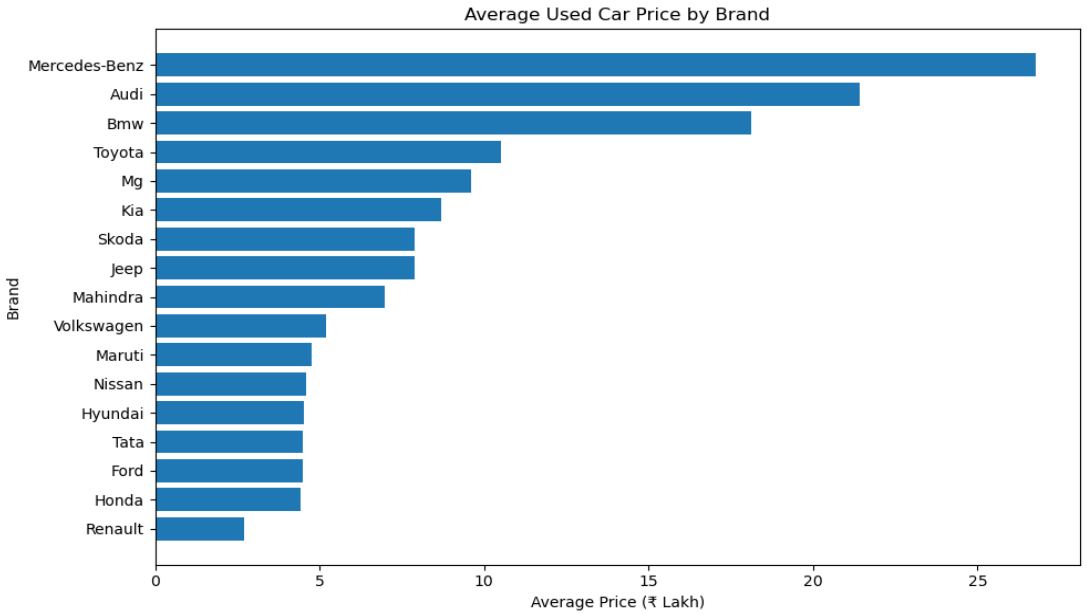
Average used-car price is ₹6.95 lakh, with a median of ₹5.68 lakh. The maximum observed price is ₹41.13 lakh.

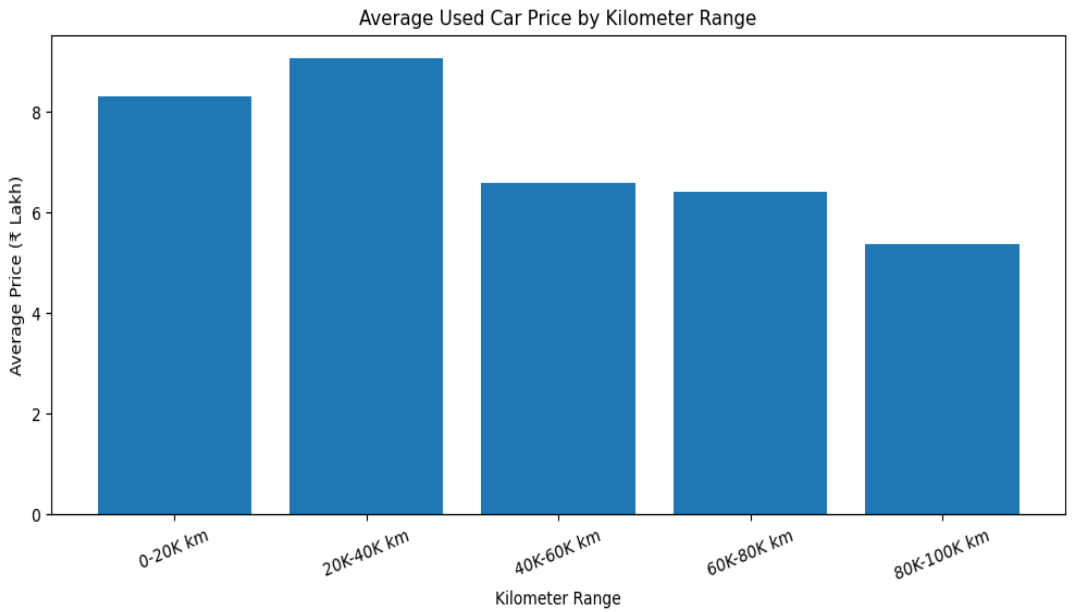
MODEL YEAR DISTRIBUTION

The dataset is concentrated around 2018–2022, with 2018 contributing 41 listings and 2021 contributing 39.

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Multivariate Analysis





BRAND-WISE AVERAGE PRICE

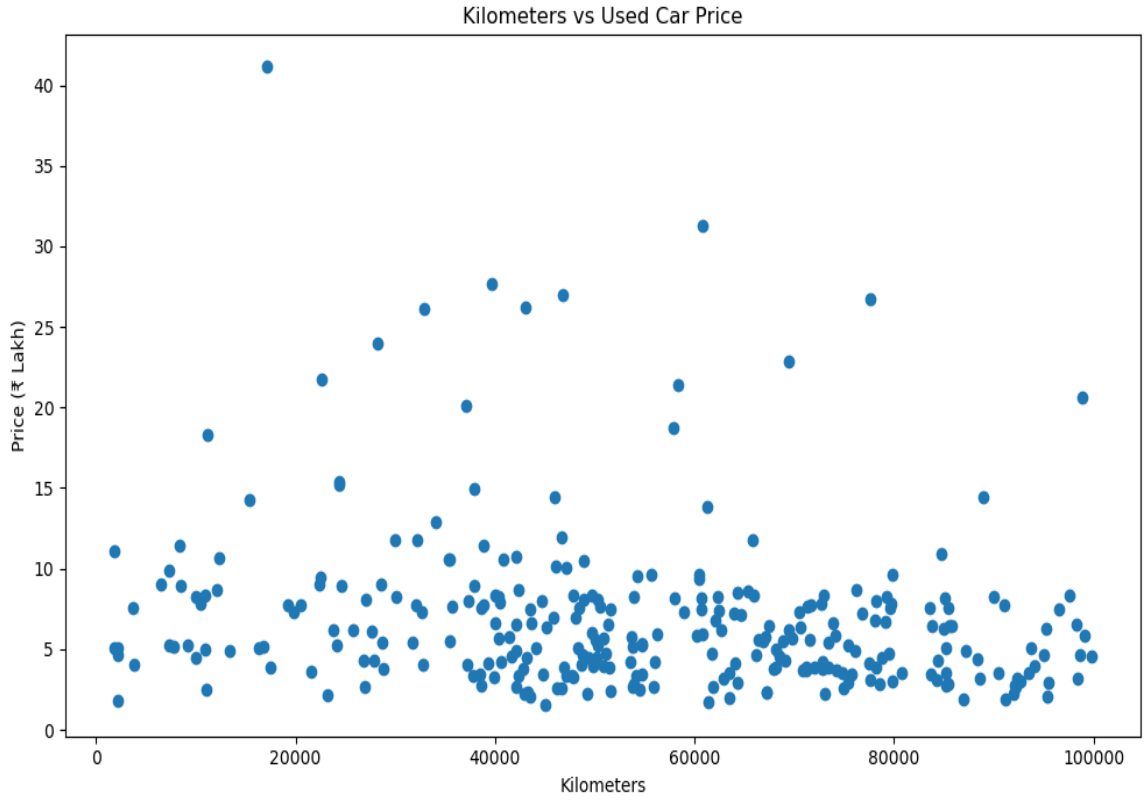
Premium brands such as Mercedes-Benz, Audi and BMW show higher average prices in the scraped sample.

KILOMETER RANGE VS PRICE

Average price is highest for cars in the 20K–40K km range and generally decreases in higher-mileage ranges.

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Correlation Analysis



KEY CORRELATIONS

- Price vs kilometers: −0.19 — weak negative relationship.
- Price vs EMI: EMI is derived from listing information and varies with price.
- Brand, fuel type and transmission show visible differences in average prices.

Correlation values describe relationships in this scraped sample; they do not prove causation.

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Statistical & Descriptive Analysis

PRICE SUMMARY

Minimum: ₹1.58 lakh

Average: ₹6.95 lakh

Median: ₹5.68 lakh

75th percentile: ₹8.13 lakh

Maximum: ₹41.13 lakh

OUTLIER ANALYSIS

IQR method applied to Price_Lakh

Q1 = 3.85 lakh

Q3 = 8.13 lakh

IQR = 4.28 lakh

Upper limit = 14.55 lakh

Price outliers detected: 18

INTERPRETATION

The market contains a broad range of used-car prices. High-value vehicles create a right-skewed distribution and should be considered separately when comparing typical prices.

Price statistics and outlier checks

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Data Quality & Validation

DATA QUALITY CHECK

- Dataset shape after cleaning: 293 × 10
- Duplicate rows: 0
- Missing values: 1 in EMI_Amount
- All other tracked columns have 0 missing values
- Model_Year stored as int64
- Price_Lakh stored as float64

KEY DISTRIBUTIONS

- Petrol: 219 | Diesel: 60 | CNG: 14
- Manual: 237 | Auto: 56
- 17 extracted brands
- 14 distinct model years
- 18 price outliers identified through IQR

Validation completed

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Key Business Insights

- The scraped Cars24 sample contains 293 used-car listings across 17 brands.
- Petrol vehicles dominate the dataset, while manual transmission is much more common than automatic.
- Automatic cars have a substantially higher average price than manual cars in this sample.
- Premium brands record higher average prices, but listing counts are uneven across brands.
- Average price is ₹6.95 lakh, while the median is ₹5.68 lakh, indicating a right-skewed market.
- Kilometers and price have a weak negative correlation, so mileage alone cannot explain resale value.
- Price outliers should be handled carefully when comparing typical used-car prices.

Conclusion

This project demonstrates how web scraping and exploratory data analysis can turn Cars24 listings into useful market insights.

Price varies with brand, transmission, fuel type, model year and mileage, but no single factor explains the entire resale price. Combining descriptive statistics, visualizations and data-quality checks helps buyers compare vehicles more meaningfully and provides a foundation for deeper predictive analysis.

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THANK YOU



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